

27/03/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456



AIM - 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

PST-I

Time : 180 Mins.

Physics: Electromagnetic Waves, Ray Optics and Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor, Experimental Skills

Chemistry: Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids, Amines, Biomolecules

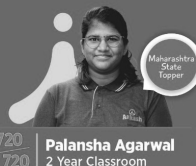
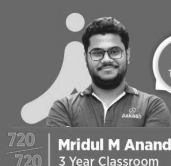
Botany: Microbes in human welfare, Organisms and populations, Molecular Basis of Inheritance, Ecosystem

Zoology: Human Reproduction, Biotechnology - Principles and Processes, Biotechnology and Its Applications

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

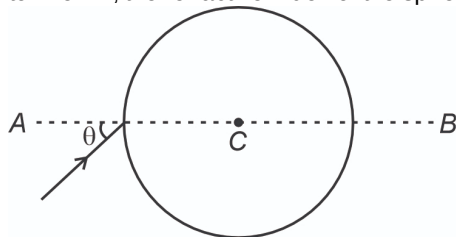
6 Aakashians scored a perfect 720/720 from Classroom Programs



PHYSICS

- In a Young's double slit experiment, if there is no initial phase difference between the light from the two slits, a point on the screen corresponding to the fifth minimum has path difference (λ is wavelength of incident light)
 - (1) $\frac{5\lambda}{2}$
 - (2) 5λ
 - (3) $\frac{9\lambda}{2}$
 - (4) $\frac{11\lambda}{2}$
- At the first minimum adjacent to the central maximum of a single slit diffraction pattern, the phase difference between the Huygens wavelet from the edge of the slit and the wavelet from the mid-point of the slit is
 - (1) π radian
 - (2) $\frac{\pi}{2}$ radian
 - (3) $\frac{\pi}{4}$ radian
 - (4) $\frac{\pi}{3}$ radian
- As the intensity of incident light of constant frequency increases, then
 - (1) Kinetic energy of emitted photoelectrons increases
 - (2) Photoelectric current decreases
 - (3) Photoelectric current increases
 - (4) Kinetic energy of emitted photoelectrons decreases
- If kinetic energy of a particle is increased to 16 times its previous value, the percentage change in the de Broglie wavelength of the particle is
 - (1) 25%
 - (2) 75%
 - (3) 50%
 - (4) 60%
- When an α -particle of mass m moving with velocity v bombards on a heavy nucleus of charge Ze , its distance of closest approach from the nucleus depends on m as
 - (1) m
 - (2) $\frac{1}{m}$
 - (3) $\frac{1}{m^2}$
 - (4) $\frac{1}{\sqrt{m}}$
- NOT gates are connected at the two inputs of a NAND gate. This combination will behave as
 - (1) NAND gate
 - (2) AND gate
 - (3) OR gate
 - (4) NOR gate
- Hydrogen atom in ground state is excited by a monochromatic radiation of $\lambda = 975 \text{ \AA}$. Number of spectral lines in the resulting spectrum emitted will be
 - (1) 3
 - (2) 2
 - (3) 6
 - (4) 10
- The de Broglie wavelength of an electron and the wavelength of a photon are the same. The ratio between the energy of that photon to the momentum of the electron is (where h is Planck's constant and c is speed of light in vacuum)
 - (1) h
 - (2) c
 - (3) $\frac{1}{h}$
 - (4) $\frac{1}{c}$
- In an interference pattern, at a point P the 16th order bright fringe or maxima for $\lambda_1 = 7200 \text{ \AA}$ is observed. What order of maxima will be visible at the point P , if the source is replaced by light of wavelength $\lambda_2 = 4800 \text{ \AA}$?
 - (1) 12
 - (2) 24
 - (3) 36
 - (4) 16
- A nucleus X undergoes following transformations
 $X \xrightarrow{\alpha} Y$ then $Y \xrightarrow{2\beta^-} Z$, then
 - (1) X and Y are isotopes
 - (2) X and Y are isobars
 - (3) X and Z are isotopes
 - (4) X and Z are isobars

11. A ray of light falls on a transparent sphere with centre C as shown in figure. If the ray emerges from the sphere parallel to line AB , the refractive index of the sphere is



- (1) $2 \sin \frac{\theta}{2}$
 (2) $\sin \frac{\theta}{2}$
 (3) $\cos \frac{\theta}{2}$
 (4) $2 \cos \frac{\theta}{2}$
12. The oscillating magnetic field in a plane EM wave is given as $B = 8 \times 10^{-6} \sin(5000\pi x - 3 \times 10^{11}\pi t)$ T. Wavelength of the wave is
- (1) 4 mm
 (2) 40 mm
 (3) 0.4 mm
 (4) 400 mm
13. An object is located at 10 cm in front of a convex lens of focal length 12 cm. Distance of image from the lens and nature of image will be
- (1) 60 cm, real
 (2) 60 cm, virtual
 (3) $\frac{60}{11}$ cm, real
 (4) $\frac{60}{11}$ cm, virtual
14. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of an hydrogen atom is
- (1) 1 : 1
 (2) 1 : -1
 (3) 2 : -1
 (4) 1 : -2
15. In two separate set up of the Young's double slit experiment, fringes of equal width are observed when light of wavelengths in the ratio 1 : 2 are used. If ratio of slit separations is 2 : 1, then ratio of the distances between the plane of the slits and the screen, in the two setup is
- (1) 4 : 1
 (2) 1 : 1
 (3) 1 : 3
 (4) 2 : 1

16. A nucleus at rest breaks into two nuclear parts which have their velocity ratio equal to 2 : 1. Ratio of their respective nuclear radii will be

- (1) $2^{1/3} : 1$
 (2) $1 : 2^{1/3}$
 (3) $2^{3/2} : 1$
 (4) $1 : 2^{3/2}$

17. Average total energy of an electromagnetic wave in vacuum is given by the relation (where E and B are amplitudes of electric and magnetic field respectively)

- (1) $\frac{1}{2} \epsilon_0 E^2 + \frac{1}{2} \mu_0 B^2$
 (2) $\frac{1}{2} \frac{E^2}{\epsilon_0} + \frac{B^2}{2\mu_0}$
 (3) $\frac{E^2 \epsilon_0 + B^2 \mu_0}{4}$
 (4) $\frac{1}{4} \epsilon_0 E^2 + \frac{B^2}{4\mu_0}$

18. A concave mirror has focal length of 50 cm. Determine the object's position to produce an inverted image that is twice the size of the object.

- (1) 75 cm
 (2) 72 cm
 (3) 63 cm
 (4) 50 cm

19. A man uses a convex lens of focal length 5 cm as magnifying glass. The least distance of distinct vision of the man is 25 cm. Find the magnification, if the image is formed at the least distance of distinct vision.

- (1) 6
 (2) 4
 (3) 3
 (4) 1.5

20. A screw gauge has a least count of 0.5 mm and its circular scale is divided into 100 equal divisions. The pitch of screw gauge is

- (1) 0.5 mm
 (2) 0.5 cm
 (3) 5 cm
 (4) 0.05 mm

21. A polariser and an analyser are inclined to each other at 45° . If the intensity of the polarised light emerging from the analyser is $4I_0$. What is the intensity of the unpolarised light incident on the polariser?

(1) $4I_0$
 (2) $8I_0$
 (3) $16I_0$
 (4) $\frac{I_0}{16}$

22. Given below are two statements, one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The focal length of a convex mirror will increase, if the mirror is placed in water.

Reason (R) : The focal length of a convex mirror of radius of curvature R is $\frac{R}{2}$.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) (A) is false but (R) is true

23. The focal length of combination of two thin lenses kept in contact is +60 cm. If the focal length of one of the lens is +25 cm, the power of the other lens will be

(1) +2.33 D
 (2) -2.33 D
 (3) -3.73 D
 (4) +3.73 D

24. Given below are two statements:

Statement I: Stability of nucleus is determined by binding energy per nucleon.

Statement II: Nuclear force acts equally between neutrons and protons.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and statement II are correct
 (2) Statement I is correct but statement II is incorrect
 (3) Statement I is incorrect but statement II is correct
 (4) Both statement I and statement II are incorrect

25. Given below are two statements, one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Hydrogen atom consists of only one electron but its emission spectrum has many lines.

Reason (R) : Lyman series constitute spectral lines corresponding to transition from higher energy to first excited state of hydrogen atom.

In the light of the above statements, choose the **correct** answer from the options given below :

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) (A) is false but (R) is true

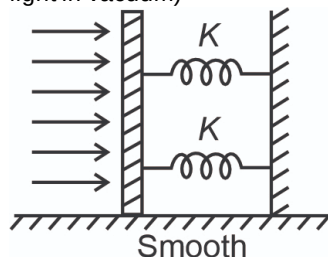
26. For photoelectric emission from certain metal the threshold frequency is ν . If radiation of frequency 2ν impinges on the metal plate, the maximum possible velocity of the emitted electron will be (m is the electron mass)

(1) $\sqrt{\frac{2h\nu}{m}}$
 (2) $2\sqrt{\frac{h\nu}{m}}$
 (3) $\sqrt{\frac{h\nu}{2m}}$
 (4) $\sqrt{\frac{h\nu}{m}}$

27. The ground state energy of H-atom is -13.6 eV. The energy needed to ionize H-atom from its second excited state is

(1) 1.51 eV
 (2) 3.4 eV
 (3) 13.6 eV
 (4) 0.85 eV

28. A perfectly reflecting plate of area A is maintained in equilibrium by striking a light of intensity I as shown. If both the springs are identical having spring constant K , then the compression in the spring at equilibrium is (c is speed of light in vacuum)



(1) $\frac{IA}{Kc}$
 (2) $\frac{2IA}{Kc}$
 (3) $\frac{IAc}{K}$
 (4) $\frac{2IAc}{K}$

29. In half wave rectification, if the input frequency is 60 Hz, then output frequency would be

- (1) 30 Hz
- (2) 60 Hz
- (3) 120 Hz
- (4) Zero

30. For the given truth table, identify the gate.

A	B	Y
1	1	0
0	1	1
1	0	1
0	0	1

- (1) NAND
- (2) XOR
- (3) NOR
- (4) OR

31. In a vernier calliper, one main scale division is x cm and $(n + 1)$ divisions of the vernier scale coincide with n divisions of the main scale. The least count (in cm) of the calliper is

- (1) $\frac{x}{n+1}$
- (2) $\left(\frac{n-1}{n}\right)x$
- (3) $\frac{x}{n-1}$
- (4) $\frac{nx}{n+1}$

32. A slit of width d is illuminated by light of wavelength 6500 Å. For what value of d will the first secondary maximum lies at angle of 30° ?

- (1) $3 \mu\text{m}$
- (2) $1.95 \mu\text{m}$
- (3) $1.55 \mu\text{m}$
- (4) $2.75 \mu\text{m}$

33. An open organ pipe has fundamental frequency of 2800 Hz. If it is dipped 20% in water, then the new fundamental frequency produced will be

- (1) 2800 Hz
- (2) 1750 Hz
- (3) 1400 Hz
- (4) 14000 Hz

34. The momentum carried by EM wave is given by

- (1) (Energy in free space) $\times$ Speed of Light
- (2) (Energy in free space) $\times \frac{1}{\text{Speed of Light}}$
- (3) (Energy in free space) $^2 \times \frac{1}{\text{Speed of Light}}$
- (4) (Energy in free space) $\times (\text{Speed of Light})^2$

35. If radius of curvature of the convex surface of a plano-convex lens is 18 cm and the refractive index of material of the lens is 1.6, the focal length of the lens is

- (1) 18 cm
- (2) 25 cm
- (3) 30 cm
- (4) 40 cm

36. Binding energy of ${}^4_2\text{He}$, ${}^7_3\text{Li}$ and ${}^{56}_{26}\text{Fe}$ are 27.37 MeV, 39.3 MeV and 492.26 MeV. Which among the three is the least stable?

- (1) ${}^4_2\text{He}$
- (2) ${}^7_3\text{Li}$
- (3) ${}^{56}_{26}\text{Fe}$
- (4) All are equally stable

37. In a meter bridge, there are two unknown resistances R_1 and R_2 . The ratio of R_1 and R_2 , if the galvanometer shows a null deflection at 20 cm from one end, is

- (1) 1 : 2
- (2) 1 : 5
- (3) 1 : 4
- (4) 2 : 3

38. The de Broglie wavelength of a gas molecule of mass m at 27°C is λ . Its wavelength at 327°C will be

- (1) $\sqrt{2}\lambda$
- (2) $\frac{\lambda}{2}$
- (3) $\frac{\lambda}{\sqrt{2}}$
- (4) 2λ

39. Energy that should be added to a freely moving electron in terms of initial energy E_1 to reduce its de Broglie wavelength from $\lambda = 10^{-10} \text{ m}$ to $0.25 \times 10^{-10} \text{ m}$

- (1) E_1
- (2) $2E_1$
- (3) $4E_1$
- (4) $15E_1$

40. Which of the following is not an example of electromagnetic wave?

- (1) Radio waves
- (2) Matter waves
- (3) X-rays
- (4) Infrared waves

41. Consider the following statements:

- (A) Convex mirrors are used as side view mirrors in automobiles because their field of view is large and produce erect and diminished images.
 (B) If the image of a real object placed near the mirror is virtual, erect and of same size, the mirror is plane.
 (C) Concave mirrors are used by dentists to view magnified image of tooth.

The correct statement(s) is/are

- (1) A and B only
- (2) B and C only
- (3) A only
- (4) All A, B and C

42. For a hydrogen atom, some of the parameters associated with electron given in column-I and their relation with principal quantum number n in column-II. Match column-I with column-II.

Column-I

- A. Radius of orbit (r)
- B. Velocity (v)
- C. Time period (T)
- D. Magnetic field produced (B) at centre

Column-II

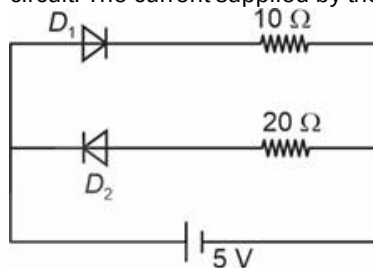
- (i) $\frac{1}{n}$
- (ii) n^2
- (iii) $\frac{1}{n^5}$
- (iv) n^3

- (1) A(ii), B(i), C(iv), D(iii)
- (2) A(i), B(ii), C(iii), D(iv)
- (3) A(i), B(iii), C(ii), D(iv)
- (4) A(iv), B(i), C(ii), D(iii)

43. In an α -decay, the kinetic energy of the daughter nucleus is 2 MeV and Q value of the reaction is 50 MeV. Assuming mother nucleus to be at rest initially, the mass number of daughter nucleus is equal to

- (1) 104
- (2) 24
- (3) 96
- (4) 120

44. Two ideal diodes are connected to a battery as shown in the circuit. The current supplied by the battery is



- (1) 0.5 A
- (2) Zero
- (3) 0.25 A
- (4) 0.75 A

45. The shortest wavelength of the Balmer series of a hydrogen-like atom (atomic number z) is the same as the shortest wavelength of the Lyman series of hydrogen atom. The value of z is

- (1) 3
- (2) 4
- (3) 2
- (4) 5

CHEMISTRY

46. Consider the following statements

- (a) Sugar moiety present in RNA molecule is β -D-ribose
- (b) Uracil is absent in DNA
- (c) Complete hydrolysis of DNA yields a hexose sugar, phosphoric acid and nitrogen containing heterocyclic compounds.

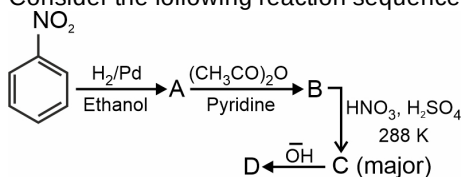
The correct statement are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

47. Which pair does not represent essential amino acid?

- (1) Histidine and Tryptophan
- (2) Lysine and Arginine
- (3) Valine and Leucine
- (4) Serine and Tyrosine

48. Consider the following reaction sequence



Product D is

- (1) CC(=O)c1ccc(N)cc1
- (2) Nc1ccc([N+](=O)[O-])cc1
- (3) Nc1ccc([N+](=O)[O-])cc1
- (4) CC(=O)Nc1ccc([N+](=O)[O-])cc1

49. The compound which does not form carbylamine on reaction with CHCl_3 and KOH is

- (1) Nc1ccccc1
- (2) Nc1ccccc1
- (3) CCNCC
- (4) CC(C)CN

50. Consider the following compounds

- (a) Benzenamine
- (b) N, N-Dimethylaniline
- (c) Phenylmethanamine

The correct order of basic strength of the given compounds in aqueous medium is

- (1) (c) > (b) > (a)
- (2) (b) > (c) > (a)
- (3) (b) > (a) > (c)
- (4) (c) > (a) > (b)

51. Given below are the two statements

Statement I: Boiling point of propan-1-ol is lower than acetone.

Statement II: Intermolecular hydrogen bonding is observed in acetone.

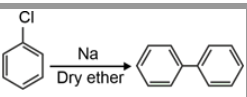
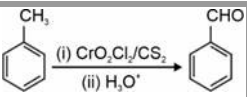
In light of above statements, choose the correct answer.

- (1) Statement I is correct but statement II is incorrect
- (2) Both statement I and statement II are correct
- (3) Both statement I and statement II are incorrect
- (4) Statement I is incorrect but statement II is correct

52. Which among the following is an acetal?

- (1) CC1(C)OC(C1)OC(C1)OC1
- (2) CCOC(=O)CC
- (3) OC1(C)OC(C1)OC(C1)OC1
- (4) C1COC(C1)OC(C1)OC1

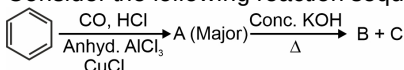
53. Match List-I with List-II.

	List I Reaction		List II Name of reaction
(a)	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2 \xrightarrow[\text{NaOH}]{\text{Br}_2} \text{R}-\text{NH}_2$	(i)	Stephen reaction
(b)		(ii)	Etard reaction
(c)		(iii)	Hoffmann bromamide degradation reaction
(d)	$\text{RCN} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) SnCl}_4/\text{HCl}} \text{RCHO}$	(iv)	Fittig reaction

The correct match is

- (1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
 (2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
 (3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
 (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

54. Consider the following reaction sequence.



Major products B and C are

- (1)  and 
 (2)  and 
 (3)  and 
 (4)  and 

55. Consider the following statements

- (a) Fehling solution B is aqueous copper sulphate.
 (b) Benzaldehyde does not give positive Fehling's test
 (c) On warming acetaldehyde with freshly prepared ammoniacal silver nitrate solution, a bright silver mirror is produced due to formation of silver metal.

The correct statements are

- (1) (a) and (b) only
 (2) (b) and (c) only
 (3) (a) and (c) only
 (4) (a), (b) and (c)

56. Given below are the two statements

Statement I: The stereoisomers related to each other as non-superimposable mirror images are called enantiomers.
 Statement II: A mixture containing two enantiomers in equal proportions will have zero optical rotation.

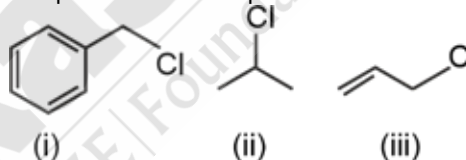
In light of above statements, choose the correct answer.

- (1) Statement I is correct but statement II is incorrect
 (2) Both statement I and statement II are correct
 (3) Both statement I and statement II are incorrect
 (4) Statement I is incorrect but statement II is correct

57. Allylic and Vinylic chloride respectively are

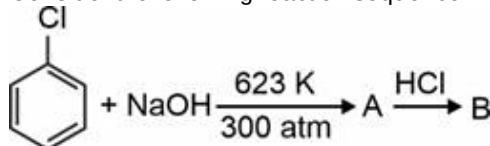
- (1)  and 
 (2)  and 
 (3)  and 
 (4)  and 

58. Correct order of rate of substitution reaction for the given compounds with KI in presence acetone is



- (1) (i) > (ii) > (iii)
 (2) (ii) > (i) > (iii)
 (3) (iii) > (ii) > (i)
 (4) (i) > (iii) > (ii).

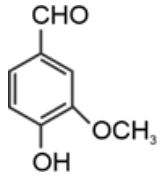
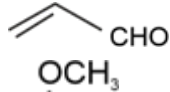
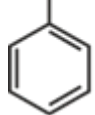
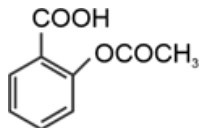
59. Consider the following reaction sequence



Incorrect statement about B is

- (1) It evolves H_2 gas on reaction with sodium
 (2) It is more acidic in nature than ethanol
 (3) It gives 2, 4, 6-tribromophenol on reaction with bromine water
 (4) On reaction with zinc dust it forms cyclohexane

60. Match List-I with List-II.

List-I Name of compound	List-II Structure
(a) Aspirin	(i) 
(b) Vanillin	(ii) 
(c) Acrolein	(iii) 
(d) Anisole	(iv) 

The correct match is

- (1) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
- (2) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
- (3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
- (4) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

61. Consider the following statements

- (a) Glucose does not give Schiff's test.
 - (b) Pentaacetate of glucose does not react with hydroxylamine.
 - (c) Glucose does not react with HCN.
- The incorrect statement(s) is/are

- (1) (a) and (b) only
- (2) (c) only
- (3) (a) and (c) only
- (4) (b) only

62. The pair of compounds which can be distinguished by iodoform test is

- (1) Benzaldehyde and Valeraldehyde
- (2) Methanol and Ethanol
- (3) Benzophenone and Isobutyraldehyde
- (4) Methyl n-propyl ketone and acetone

63. Cinnamaldehyde can be obtained by the aldol condensation reaction of

- (1) Benzaldehyde and acetone
- (2) Acetophenone and benzaldehyde
- (3) Formaldehyde and acetaldehyde
- (4) Benzaldehyde and acetaldehyde

64. Consider the following statements

- (a) Benzenediazonium chloride is insoluble in water
- (b) Benzenediazonium fluoroborate is stable at room temperature
- (c) Benzenediazonium chloride is reduced to benzene by the use of ethanol.

The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

65. Lactose is composed of

- (1) β -D-Galactose and β -D-glucose
- (2) α -D-Glucose and β -D-glucose
- (3) α -D-Glucose and β -D-fructose
- (4) α -D-Galactose and β -D-fructose

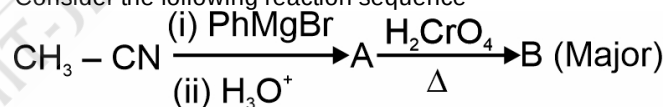
66. Match List -I with List-II.

List-I Name of vitamin	List-II Deficiency disease
(a) Vitamin E	(i) Convulsions
(b) Pyridoxine	(ii) Increased blood clotting time
(c) Riboflavin	(iii) Increased fragility of RBCs
(d) Vitamin K	(iv) Digestive disorders

The correct match is

- (1) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- (2) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (3) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
- (4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

67. Consider the following reaction sequence



Major product B is

- (1) Benzaldehyde
- (2) Benzoic acid
- (3) Benzyl alcohol
- (4) Phenol

68. Correct order of acidic strength of the given compounds is

- (i) BrCH_2COOH
 - (ii) HCOOH
 - (iii) $\text{C}_6\text{H}_5\text{COOH}$
 - (iv) $\text{ClCH}_2\text{CH}_2\text{COOH}$
- (1) (i) > (ii) > (iv) > (iii)
 - (2) (ii) > (iii) > (i) > (iv)
 - (3) (i) > (iv) > (ii) > (iii)
 - (4) (ii) > (i) > (iv) > (iii)

69. Which among the following compounds gives alkene as major product on reaction with copper at 573 K?

- (1)
- (2)
- (3)
- (4)

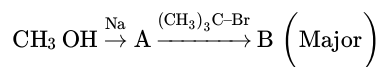
70. Total number of compounds (including stereoisomers) obtained on monochlorination of 2-methylpentane is

- (1) 9
- (2) 8
- (3) 6
- (4) 7

71. Major product (A) formed in given reaction is

-
- (1)
 - (2)
 - (3)
 - (4)

72. Consider the following reaction sequence



Major product B is

- (1)
- (2)
- (3)
- (4)

73. Incorrect statement about hormones is

- (1) Hormones are molecules that act as intercellular messengers
- (2) Glucagon tends to increase the glucose level in the blood
- (3) Thyroxine is an iodinated derivative of amino acid lysine
- (4) Progesterone is responsible for preparing the uterus for implantation of fertilised egg

74. Which among the following esters gets hydrolysed most easily under alkaline conditions?

- (1)
- (2)
- (3)
- (4)

75. Consider the following name reactions

- Wurtz reaction
- Finkelstein reaction
- Swarts reaction
- Gatterman-Koch reaction

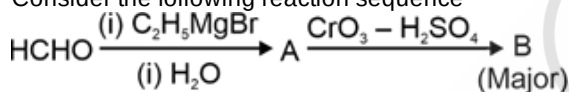
The reactions in which C–C bond formation take place are

- (a) and (b) only
- (a), (b) and (d) only
- (a) and (d) only
- (b), (c) and (d) only

76. Phenetole on reaction with HI majorly gives

-  and $\text{CH}_3 - \text{I}$
-  and $\text{CH}_3\text{CH}_2 - \text{I}$
-  and $\text{CH}_3 - \text{I}$
-  and $\text{CH}_3\text{CH}_2 - \text{I}$

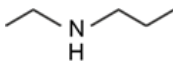
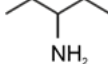
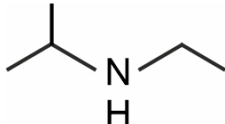
77. Consider the following reaction sequence



Major product B is

- $\text{CH}_3\text{CH}_2\text{CHO}$
- $\text{CH}_3\text{CH}_2\text{COOH}$
- $\text{CH}_3\text{CH} = \text{CH}_2$
- $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_3$

78. A compound 'X' having molecular formula $\text{C}_5\text{H}_{13}\text{N}$ on reaction with Hinsberg's reagent gives compound 'Y' which dissolves in aqueous alkali. The structure of the compound 'X' could be

- 
- 
- 
- 

79. Given below are the two statements

Statements I: Amylose is water soluble component of starch.

Statement II: Amylose is a branched chain polymer of α -D-glucose units.

In light of above statements, choose the correct answer.

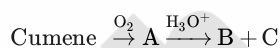
- Statement I is correct but statement II is incorrect
- Statement I is incorrect but statement II is correct
- Both statement I and statement II are correct
- Both statement I and statement II are incorrect

80. Correct order of boiling point of the given compounds is

- $\text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2$
- $(\text{C}_2\text{H}_5)_2\text{NH}$
- $n\text{-C}_4\text{H}_9\text{NH}_2$

- (i) > (iii) > (ii)
- (iii) > (ii) > (i)
- (iii) > (i) > (ii)
- (ii) > (i) > (iii)

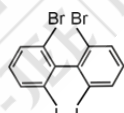
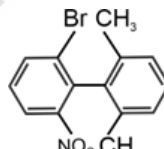
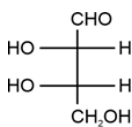
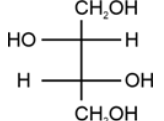
81. Consider the following reaction sequence



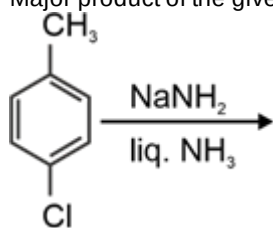
Major products B and C are

- Phenol and acetone
- Benzyl alcohol and acetaldehyde
- Phenol and propanal
- Benzyl alcohol and ethyl alcohol

82. Optically inactive compound among the following is

- 
- 
- 
- 

83. Major product of the given reaction is



- (1)
- (2)
- (3)
- (4)

84. Correct nucleophilicity order of the given species in ethanol is

- (1) $\text{EtO}^- > \text{I}^- > \text{PhO}^-$
- (2) $\text{I}^- > \text{EtO}^- > \text{PhO}^-$
- (3) $\text{I}^- > \text{PhO}^- > \text{EtO}^-$
- (4) $\text{EtO}^- > \text{PhO}^- > \text{I}^-$

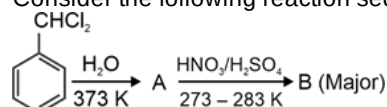
85. Match List-I with List-II

- | List-I
(Carbonyl derivative) | List-II
(Name) |
|---------------------------------|--------------------|
| (a) $\text{>C=N-NH-C(=O)-NH}_2$ | (i) Hydrazone |
| (b) >C=NH | (ii) Semicarbazone |
| (c) >C=N-OH | (iii) Imine |
| (d) >C=N-NH_2 | (iv) Oxime |

The correct match is

- (1) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
- (2) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)
- (3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
- (4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

86. Consider the following reaction sequence



Major product B is

- (1)
- (2)
- (3)
- (4)

87. Consider the following statements

- (a) Carboxylic acid are higher boiling liquids than aldehydes or ketones of comparable molecular masses.
- (b) Acetic acid liberates CO_2 on reaction with NaHCO_3 .
- (c) Picric acid does not liberate CO_2 on reaction with NaHCO_3 .

The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

88. Given below are the two statements:

One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): Aromatic carboxylic acids do not undergo Friedel-Crafts reaction.

Reason (R): Carboxyl group in aromatic carboxylic acid is deactivating in nature and the catalyst aluminium chloride gets bonded to the carboxyl group.

In light of above statements, choose the correct answer from the options given below.

- (1) Both A and R are true but R is not the correct explanation of A
- (2) A is true but R is false
- (3) A is false but R is true
- (4) Both A and R are true and R is correct explanation of A

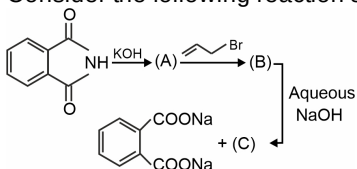
89. Consider the following compounds

- (a) Isopropyl alcohol
- (b) Isobutyl alcohol
- (c) Sec-butyl alcohol
- (d) n-propyl alcohol

The compounds which will give blue colour in Victor Meyer's test are

- (1) (a), (b) and (d) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (b) and (d) only

90. Consider the following reaction sequence



Product C is

- (1) CH2=CH-CH2OH
- (2) CH2=CH-C(=O)NH2
- (3) CH2=CH-CH2NH2
- (4) CH2=CH-CHO

BOTANY

91. Interaction between which of the following organisms does **not** fall under the predator-prey relationship?

- (1) Sparrow and seeds
- (2) Prickly pear cactus and moth
- (3) *Cuscuta* and hedge plants
- (4) Grass and goat

92. Two bird species are competing for the same tree fruit in a garden. To survive and coexist together, they may avoid competition in the same garden by

- (1) Predating on each other
- (2) Increasing the time spent on attacking each other
- (3) Choosing different foraging pattern
- (4) Feeding at the same time

93. If there are 850 snails in a pond, and within a year their number increase to 8500 by reproduction. What should be their birth rates per snail per year?

- (1) 11
- (2) 0.11
- (3) 9
- (4) 10

94. Beetles with red and black markings, the ladybirds, are useful to get rid of

- (1) Mosquitoes
- (2) Aphids
- (3) Butterfly caterpillars
- (4) *Trichoderma*

95. All of the following are true for penicillin, **except**

- (1) It was discovered while working on *Streptococcus* bacteria
- (2) It was named after the mould, *Penicillium notatum*
- (3) It was extensively used to treat the American soldiers wounded in World War II
- (4) Its full potential as an antibiotic was established by Ernest Chain and Howard Florey

96. Which of the following is **not** true with respect to LAB?

- (1) Acids produced by LAB completely digest the milk proteins
- (2) A small amount of curd contains millions of LAB
- (3) They multiply at suitable temperatures
- (4) They convert milk to curd, increasing its nutritional quality

97. Which of the following organisms is responsible for the production of large amount of CO₂ during the ripening of Swiss cheese?

- (1) *Saccharomyces cerevisiae*
- (2) *Propionibacterium sharmanii*
- (3) Lactic acid bacteria (LAB)
- (4) *Aspergillus niger*

98. From the options given below, identify the organism which is used for producing statins.

- (1) *Trichoderma polysporum*
- (2) *Monascus purpureus*
- (3) *Streptococcus*
- (4) *Acetobacter aceti*

99. Regarding the types of ecosystems, pick the **odd** one out from the following.

- (1) Estuaries
- (2) Pond
- (3) Forest
- (4) Wetland

100. Which of the following is **not** a characteristic of anthropogenic ecosystem?

- (1) Shows little diversity
- (2) Possesses self-regulatory mechanism
- (3) Has high productivity
- (4) Forms a simple food chain

101. Soluble inorganic nutrients get precipitated as unavailable salts during which of the following steps of detritus decomposition?

- (1) Catabolism
- (2) Fragmentation
- (3) Leaching
- (4) Humification

102. Following are given some ecosystems.

- (A) Tree ecosystem
- (B) Grassland ecosystem
- (C) Pond ecosystem
- (D) Forest ecosystem

In how many of the above mentioned ecosystems, GFC will act as the major conduit of energy?

- (1) One
- (2) Two
- (3) Three
- (4) Four

103. Which of the following statements is **incorrect**?

Solar input, temperature, day length and other climatic conditions regulate the rate of functions of an aquatic ecosystem, such as pond.

- (1) Phytoplanktons, some algae and floating submerged plants act as producers in a pond ecosystem.
- (2) Fungi, bacteria and flagellates are the decomposers in a pond.

All the components of ecosystem are seen to function in a pond which eventually leads to the unidirectional movement of energy from higher to the lower trophic levels.

- (3) Fungi, bacteria and flagellates are the decomposers in a pond.
- (4) All the components of ecosystem are seen to function in a pond which eventually leads to the unidirectional movement of energy from higher to the lower trophic levels.

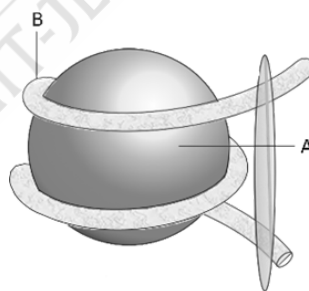
104. Mortality refers to the number of individuals that

- (1) Have died during a given period in a population
- (2) Have entered the population during a given period
- (3) Are born during a given period in a population
- (4) Have left the habitat and gone elsewhere during a given period

105. Which of the following is **not** true w.r.t. salient features of our genetic material?

- (1) It has a backbone of sugar-phosphate, and the bases project inside
- (2) The two polynucleotide chains run in anti-parallel fashion
- (3) The bases of two strands are paired through glycosidic bonds
- (4) The plane of one base pair stacks over the other in double helix

106. Observe the figure given below and choose the **correct** option to label **A** and **B**.



- (1) **A** – DNA, **B** – Histone octamer
- (2) **A** – H₁ histone, **B** – DNA
- (3) **A** – Histone octamer, **B** – DNA
- (4) **A** – H₁ histone, **B** – Histone octamer

- 107.** "The ratios between Adenine and Thymine and Guanine and Cytosine are constant and equals one." This observation was done by :
- (1) Maurice Wilkins
 - (2) Rosalind Franklin
 - (3) Erwin Chargaff
 - (4) Frederick Griffith
- 108.** Select the **odd** one with respect to the structure of a polynucleotide chain.
- (1) Cytosine is common for the both DNA and RNA
 - (2) Uracil is present in RNA at the place of Guanine
 - (3) A phosphate group is linked to OH of 5'C of a nucleoside through phosphoester linkage
 - (4) A free OH is present towards the 3'-end of the chain
- 109.** Identify the **incorrect** one w.r.t. Griffith's experiment?
- (1) Mice injected with heat-killed S strain bacteria died due to pneumonia infection
 - (2) Mice injected with R strain bacteria lived
 - (3) Mice injected with live S strain bacteria died due to pneumonia infection
 - (4) Mice injected with live R strain + heat-killed S strain bacteria died due to pneumonia infection
- 110.** All of the following are true for RNA, **except**
- (1) It was the first genetic material
 - (2) It acts as a catalyst in some biochemical reactions
 - (3) It is reactive and unstable
 - (4) It mutates slower than DNA
- 111.** In the structural gene of a transcription unit, the template strand
- (1) Has the polarity 5' → 3'
 - (2) Is the non-coding strand
 - (3) Is also called as the sense strand
 - (4) Has the promoter sequence towards its 5' end
- 112.** Meselson and Stahl conducted the experiment to prove the semi-conservative nature of DNA. The DNA that was extracted from the 1st generation of *E. coli* of this experiment
- (1) Did not show the presence of hybrid DNA
 - (2) Had an intermediate density
 - (3) Was composed of equal amounts of hybrid and light DNA
 - (4) Had the maximum density
- 113.** In eukaryotes, the regulation of gene expression may occur
- (a) at transcriptional level.
 - (b) during the regulation of splicing.
 - (c) translational level.
- The **correct** ones are
- (1) (a) and (c) only
 - (2) All (a), (b) and (c)
 - (3) (b) and (c) only
 - (4) (a) and (b) only
- 114.** Which among the following statements is **false** for the process of transcription in bacteria?
- (1) It can be coupled with translation
 - (2) It is followed by translation in a separate compartment
 - (3) It does not require any processing of mRNA
 - (4) It uses single DNA-dependent RNA polymerase to catalyse the transcription of all types of RNA
- 115.** The chemical method which was instrumental in synthesising RNA molecules with defined combinations of bases was developed by
- (1) George Gamow
 - (2) Har Gobind Khorana
 - (3) Meselson and Stahl
 - (4) Jacob Monod
- 116.** Identify the **correct** sequence on the mRNA strand if the sequence on the corresponding template strand is – 3'TAGCTAGCTAGCTAG 5'
- (1) 5'AUCAGUCAGUCGAUC 3'
 - (2) 3'AUCGAUCGAUCGAUC 5'
 - (3) 5'AUCGAUCGAUCGAUC 3'
 - (4) 5'AUGUAGUAAUAGGUC 3'
- 117.** Select the **mismatched** pair from the following.
- (1) SNPs – Single Nucleoside Polymorphisms
 - (2) YAC – Yeast Artificial Chromosome
 - (3) BAC – Bacterial Artificial Chromosome
 - (4) ESTs – Expressed Sequence Tags
- 118.** There are some salient observations drawn from human genome project. Which amongst the following statements regarding the observations is **true**?
- (1) Chromosome X has most gene
 - (2) Only 12 percent of the genome codes for protein
 - (3) The average gene consists of 3000 bases
 - (4) Snips occur at about 4.1 million locations

119. Following are two statements labelled as **Assertion (A)** and **Reason (R)**.

Assertion (A): Life cycle of endoparasites is complex in nature.

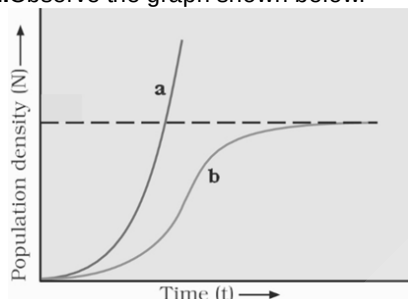
Reason (R): Morphological and anatomical features of endoparasites are greatly simplified while emphasizing their reproductive potential.

- (1) (A) is true but (R) is false
- (2) (A) is false but (R) is true
- (3) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (4) Both (A) and (R) are true and (R) is the correct explanation of (A)

120. In the *lac* operon, gene 'z' codes for

- (1) Permease
- (2) Transacetylase
- (3) Repressor protein
- (4) Beta-galactosidase

121. Observe the graph shown below.



Identify the **correct** equation for the curve 'a'.

- (1) $\frac{dN}{dt} = rN \left(\frac{K-N}{K} \right)$
- (2) $\frac{dN}{dt} = rN$
- (3) $N_{(t+1)} = N_t + \left[(B + I) - (D + E) \right]$
- (4) $\frac{dN}{dt} = rN \left(\frac{K+N}{K} \right)$

122. The population age pyramids for growing and mature populations are respectively,

- (1) Bell-shaped and urn-shaped
- (2) Triangular and bell-shaped
- (3) Urn-shaped and triangular
- (4) Bell-shaped and triangular

123. Match the **Column I** with **Column II** and select the correct option.

Column I		Column II	
a.	Sea anemone and clown fish	(i)	Parasitism
b.	Mycorrhizae	(ii)	Competition
c.	Marine fish and copepods	(iii)	Mutualism
d.	Abingdon tortoise and goat	(iv)	Commensalism

- (1) a(iv), b(iii), c(ii), d(i)
- (2) a(i), b(iv), c(ii), d(iii)
- (3) a(iv), b(iii), c(i), d(ii)
- (4) a(iii), b(ii), c(iv), d(i)

124. Read the statements given below.

(A) *Bacillus thuringiensis*, TMV and aphids are the biocontrol agents.

(B) *Anabaena* is an N_2 -fixing bacteria which acts as a biofertilizer in the paddy fields.

(C) Certain anaerobic bacteria are involved in the production of biogas.

(D) Baculoviruses are the excellent candidates for broad spectrum insecticidal applications.

Which of the above statements are **false**?

- (1) (A) and (B) only
- (2) (A) and (D) only
- (3) (B), (C) and (D) only
- (4) (A), (B) and (D) only

125. Tertiary treatment of the sewage involves

- (1) Sequential filtration
- (2) Sedimentation
- (3) Formation of activated sludge
- (4) Reverse osmosis

126. Read the statements given below.

(A) Sun is the only source of energy for all the ecosystems on Earth.

(B) Only 1-5 percent of the total incident solar radiations are photosynthetically active radiation.

(C) Plants and photosynthetic bacteria fix Sun's radiant energy to make food from simple inorganic material.

(D) Ecosystems are not exempt from the 2nd law of thermodynamics.

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) (A), (B) and (D) are incorrect
- (2) (A), (B) and (C) are correct
- (3) (C) and (D) are correct
- (4) (B) and (D) are correct

127. X is not considered as a functional aspect of an ecosystem.
Choose the **correct** option for X.
- (1) Productivity
 - (2) Energy flow
 - (3) Nutrient cycling
 - (4) Stratification

128. If 60 J energy of the incident solar radiation is trapped at the producer level, then the energy available to the T₄ level in the following food chain will be
Grass → Grasshopper → Frog → Snake
- (1) 6 J
 - (2) 0.006 J
 - (3) 0.06 J
 - (4) 0.6 J

129. Following are given two statements, labelled as **Assertion (A)** and **Reason (R)**.
Assertion (A): We can predict the base sequence of the other strand of DNA if the sequence of one strand is known.
Reason (R): Both the strands of DNA are complementary to each other.
In the light of the above statements, choose the **correct** option.
- (1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
 - (2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**
 - (3) **(A)** is true but **(R)** is false
 - (4) **(A)** is false but **(R)** is true

130. Histones in eukaryotes
- a. Are rich in lysine and arginine.
 - b. Are the sets of negatively charged proteins.
 - c. Are organised to form a unit of eight molecules.
 - d. Are majorly composed of acidic proteins.
- The **correct** ones are
- (1) a, b and d only
 - (2) a and c only
 - (3) b, c and d only
 - (4) a, c and d only

131. Identify the **correct** function of the operator gene from the following.
- (1) Synthesises a protein which can act as an activator or a repressor
 - (2) Provides attachment site for RNA polymerase
 - (3) Interacts with the product of regulator gene
 - (4) Catalyzes the transcription of mRNA for polypeptide synthesis

132. When viewed under electron microscope, the nucleosomes in chromatin are seen as
- (1) Histone tetramers
 - (2) Double helix structure
 - (3) Beads on string
 - (4) Clover leaf like structure

133. The term 'Nuclein' was given by
- (1) Altmann
 - (2) Friedrich Meischer
 - (3) Maurice Wilkins
 - (4) James Watson

134. During the process of transcription RNA polymerase II transcribes
- (1) snRNA and tRNA
 - (2) hnRNA
 - (3) rRNA
 - (4) 5S rRNA

135. With respect to predation, choose the **incorrect** statement from the following.
- (1) Predators act as conduits of energy transfer across the trophic levels
 - (2) Agricultural pest control are based on the ability of the predator to regulate prey population
 - (3) Both the interacting species are negatively affected
 - (4) Might lead to species extinction when predator overexploits its prey

ZOOLOGY

136. Presently, how many recombinant therapeutics have been approved for human use all over the world?
- (1) 40
 - (2) 12
 - (3) 24
 - (4) 30

137. In human females, the primary sex organ that produces the female gamete and several steroidal hormones is
- (1) Ovary
 - (2) Uterus
 - (3) Mammary gland
 - (4) Vagina

- 138.** Listed below are some of the components of the human female mammary gland:
(a) Mammary duct
(b) Lactiferous duct
(c) Alveoli
(d) Ampulla
Among these, choose the one that represents the most distal part.
- (1) (a)
(2) (b)
(3) (c)
(4) (d)
- 139.** Select the odd one w.r.t. transgenic animals/ products.
- (1) Golden rice
(2) Rosie
(3) Pomato
(4) Bt brinjal
- 140.** A circular DNA has two recognition sites for *EcoRI*. How many fragment(s) will be obtained after complete digestion of this circular DNA with *EcoRI*?
- (1) Three
(2) Two
(3) One
(4) Four
- 141.** Which among the following is a nematode which infects the roots of tobacco plants and causes a great reduction in their yield?
- (1) *E. coli*
(2) *Haemophilus influenzae*
(3) *Meloidogyne incognita*
(4) *Agrobacterium tumefaciens*
- 142.** Select the odd one w.r.t. androgens in males.
- (1) Steroidal in nature
(2) Regulate the development of prostate gland
(3) Are released by the cells which are present in the interstitial spaces of testes
(4) Increase the pitch of the voice
- 143.** The construction of the first recombinant DNA emerged from the possibility of linking gene encoding antibiotic resistance with a native plasmid of
- (1) *Salmonella typhimurium*
(2) *Escherichia coli*
(3) *Agrobacterium tumefaciens*
(4) *Bacillus thuringiensis*
- 144.** The group of enzymes which remove nucleotides from the ends of the DNA belong to which class of enzymes?
- (1) Class VI
(2) Class V
(3) Class III
(4) Class II
- 145.** • A vector-less method of gene transfer in which the recombinant DNA is directly injected into the nucleus of a mice cell is called
- (1) Micro-injection
(2) Gene gun
(3) Biolistics
(4) Heat shock method
- 146.** Read the following statements.
Statement A: Today, we know more than 900 restriction enzymes isolated from over 20 strains of eukaryotes, each of which recognises the same recognition sequence.
Statement B: If a 3 kb dsDNA undergoes 30 cycles of PCR, then 1 billion copies of the original template strands will be obtained.
Choose the correct option.
- (1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct
- 147.** Select the correct match w.r.t. the location of different structures in human females.
- (1) Areola - Present around oviduct
(2) Clitoris - Present above urethral opening
(3) Hymen - Covers the opening of cervix
(4) Fimbriae - Part of uterus
- 148.** How many spermatids will be obtained from 5 primary spermatocytes after the completion of meiosis?
- (1) 10
(2) 40
(3) 60
(4) 20
- 149.** In humans, after implantation, finger-like projections appear on the trophoblast called
- (1) Chorionic villi
(2) Inner cell mass
(3) Ectoderm
(4) Fimbriae

- 150.** Choose the correct option to complete the analogy.
Lepidopterans : Army worms :: Dipterans _____
- (1) Mosquitoes
 - (2) Beetles
 - (3) Tobacco budworms
 - (4) Cotton bollworms
- 151.** The process by which isolated protoplasts from two different varieties of plants can be fused to form a hybrid protoplast which can be further grown to form a new plant is called
- (1) Micro-propagation
 - (2) Somatic hybridisation
 - (3) Recombinant DNA technology
 - (4) Gene therapy
- 152.** Observe the flowchart given below w.r.t. hormonal control of spermatogenesis in humans.
- Select the option that correctly identifies A, B and C respectively.
- (1) Estrogen, FSH, Sustentacular cells
 - (2) FSH, LH, Leydig cells
 - (3) LH, FSH, Leydig cells
 - (4) FSH, Relaxin, Sustentacular cells
- 153.** Read the following statements w.r.t. restriction endonucleases:
- (a) Each of them functions by inspecting the full length of a DNA sequence.
 - (b) Each of them recognises a specific sequence of base pairs in DNA that reads the same on the two strands when the orientation of reading is kept the same.
 - (c) These enzymes always cut the strands of DNA far away from the centre of palindromic sites.
 - (d) These enzymes specifically cut the hydrogen bonds only to cleave the desired DNA fragment.
- How many of the above given statements is/are **incorrect** w.r.t. them?
- (1) One
 - (2) Two
 - (3) Three
 - (4) Four
- 154.** The human protein-enriched milk produced by the first transgenic cow was/contained
- (1) Human α -1-antitrypsin
 - (2) Nutritionally less balanced product for human babies than natural cow milk
 - (3) 24 grams of human protein per litre
 - (4) Human alpha-lactalbumin
- 155.** In a hypothetical condition, if a 5-weeks pregnant female consumes a drug which inhibits the secretion of hCG, then
- (1) Pregnancy will still continue normally
 - (2) Corpus luteum will still continue secreting progesterone at normal levels
 - (3) Placenta will start secreting high levels of progesterone immediately for compensating the loss of hCG
 - (4) Endometrium will start shedding off due to decrease in progesterone levels
- 156.** Permanent cure for a child suffering from ADA deficiency could be
- (1) Bone marrow transplantation after birth, before the onset of symptoms
 - (2) Enzyme replacement therapy
 - (3) Gene therapy after the onset of symptoms
 - (4) Introduction of functional ADA gene into the cells at the early embryonic stages
- 157.** In the menstrual cycle of a human female, the secretory phase coincides with the phase in which
- (1) Growth of follicles occurs in ovaries
 - (2) Menstrual flow occurs
 - (3) Corpus luteum secretes progesterone and estrogen
 - (4) The gonadotropin surge induces ovulation
- 158.** If a foreign DNA is inserted at the *Sal* I site in vector pBR322, then after transformation in *E.coli*
- (1) All the recombinants will grow on tetracycline containing medium
 - (2) All the non-recombinants will not grow on ampicillin containing medium
 - (3) All recombinant transformants will grow only on ampicillin containing medium
 - (4) All non-transformants will grow on tetracycline containing medium
- 159.** Consider the following w.r.t. *Eco*RI :
- (a) Isolated from bacteria that belong to the species *Escherichia*
 - (b) Recognises a recognition sequence of six base pairs
 - (c) Produces staggered ends
- Choose the option which represents only the correct features.
- (1) (b) and (c) only
 - (2) (a), (b) and (c)
 - (3) (a) and (b) only
 - (4) (a) and (c) only

160.After gel electrophoresis, the separated bands of DNA are cut out and extracted from the gel piece in a step known as

- (1) Transformation
- (2) Biolistics
- (3) Electroporation
- (4) Elution

161.The thermostable DNA polymerase utilised in PCR technique is isolated from

- (1) *Bacillus thuringiensis*
- (2) *Thermus aquaticus*
- (3) *Salmonella typhimurium*
- (4) *Meloidogyne incognita*

162.For bacterial cells to take up DNA, they must first be made competent, thus they are treated with

- (1) Armed pathogen vectors
- (2) Low velocity microparticles of gold
- (3) DNA of *Agrobacterium tumefaciens*
- (4) Specific concentration of divalent cations

163.In humans, the formation of the first polar body is seen in a follicle which is characterised by the presence of

- (1) Only a single layer of granulosa cells
- (2) Fluid-filled cavity
- (3) Yellow-coloured temporary endocrine structure
- (4) Undifferentiated theca layer

164.Match column I with column II and choose the correct option.

Column I	Column II
(a) Biolistics	(i) Artificial plasmid vector
(b) pBR322	(ii) Vector-less method of gene transfer
(c) β - galactosidase	(iii) Made from synthetic monomers
(d) Agarose gel	(iv) Reporter enzyme
	(v) Made from natural polymers from sea weeds

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(v), b(ii), c(i), d(iv)
- (3) a(ii), b(i), c(iv), d(v)
- (4) a(ii), b(i), c(iv), d(iii)

165.Once ingested by an insect, Bt-toxin is converted into an active form due to the

- (1) Acidic pH of its foregut
- (2) Neutral pH of its hindgut
- (3) Alkaline pH of its gut
- (4) Acidic pH of saliva in its mouth

166.The cells harbouring the cloned gene of interest can be multiplied in a continuous culture system where

- (1) High yields of desired protein are obtained under anaerobic conditions only
- (2) Cells are maintained in their physiologically most active exponential phase
- (3) Fresh medium is added only once at the start
- (4) Cells are maintained in their physiologically stationary phase

167.A sequence from where replication starts and any piece of DNA when linked to this, can be made to replicate within the host cell, is known as

- (1) Selectable marker
- (2) Recognition site
- (3) Cloning site
- (4) Origin of replication

168.In human sperm, the _____ is filled with enzymes that help in fertilisation of the ovum. Choose the correct option to fill in the blank.

- (1) Acrosome
- (2) Neck
- (3) Middle piece
- (4) Tail

169.If an alien gene encoding for β -galactosidase is ligated at *Hind* II in pBR322, then after complete transformation, in the presence of chromogenic substrate

- (1) Transformants will appear white and show sensitivity towards ampicillin
- (2) Recombinants will produce blue coloured colonies
- (3) Non-recombinants will produce blue coloured colonies
- (4) Recombinants will be *amp*^S and *tet*^R

170.Assertion (A): Insulin is always injected, never administered orally to diabetic patients.

Reason (R): Insulin is a peptide hormone which can be digested in alimentary canal due to action of digestive enzymes.

In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

171. A probe can be a

- (1) dsRNA tagged with a radioactive molecule
- (2) ssRNA only, tagged with a non-radioactive molecule
- (3) ssRNA or ssDNA tagged with a radioactive molecule
- (4) Circular double-stranded DNA tagged with a radioactive molecule

172. The basic step(s) required for genetically modifying an organism are

- (a) Identification of DNA with desirable genes
 - (b) Introduction of the identified DNA into the host
 - (c) Maintenance of introduced DNA in the host and transfer of the DNA to its progeny
- Select the **correct** option.

- (1) Only (a)
- (2) (a) and (b) only
- (3) (b) and (c) only
- (4) (a), (b) and (c)

173. If a human male ejaculates 300 million sperms per ejaculate, then for normal fertility, at least how many of them must have normal size, shape and show vigorous motility?

- (1) 72 million
- (2) 200 million
- (3) 240 million
- (4) 160 million

174. **Assertion (A):** In all male mammals, ureters act as the urinogenital duct.

Reason (R): In all male mammals, ureters transport both urine as well as sperms to the outside of the body.
In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

175. Select the **incorrect** statement w.r.t. gel electrophoresis.

- (1) Commonly used matrix is a natural polymer extracted from sea weeds
- (2) DNA fragments separate according to their size through the sieving effect provided by agarose gel
- (3) The smaller the fragment size, the farther it moves
- (4) DNA fragments can be separated by forcing them to move towards cathode under the effect of electric field.

176. If a DNA fragment is obtained after the action of restriction enzyme *Cla* I, then for its successful ligation, cloning vector has to be cleaved by using

- (1) *Eco* RI
- (2) *Pvu* II
- (3) *Cla* I
- (4) *Bam* HI

177. Select the incorrect statement w.r.t. humans.

- (1) By the end of first trimester, the body of the foetus is covered with fine hair.
- (2) Inner cell mass of the embryo contains the stem cells.
- (3) Prolactin has no role in the foetal-ejection reflex.
- (4) Fructose present in the seminal fluid is the major source of energy for the movement of spermatozoa.

178. *E. coli* defends itself from the attack of bacteriophages by

- (1) Producing restriction exonucleases that cut the genome of the bacteriophages
- (2) Producing restriction enzymes that cut the DNA of bacteriophages
- (3) Adding methyl groups to adenine and cytosine bases of the bacteriophage's DNA
- (4) By using alkaline phosphatase to prevent the attachment of bacteriophages

179. How many of the restriction sites given in the box below are present within the selectable marker genes in vector pBR322?

Pst I, *Pvu* I, *Bam* HI, *Pvu* II, *Eco* RI

Select the correct option.

- (1) Three
- (2) Two
- (3) Four
- (4) Five

180. Read the following statements :

Statement (A): Formation of haploid second polar body occurs in fallopian tube.

Statement (B): Entry of sperm into the cytoplasm of ovum induces the completion of second meiotic division of the diploid secondary oocyte resulting in formation of a haploid ootid and second polar body.

Choose the correct option w.r.t. humans.

- (1) Both statements (A) and (B) are correct
- (2) Both statements (A) and (B) are incorrect
- (3) Only statement (B) is correct
- (4) Only statement (B) is incorrect



**Scan the QR Code to know
"How can you score 720 with AIM 720"**

